



Akoma Ntoso for EU (AKN4EU) documentation

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Volume I – General principles

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Executive summary

The EU Institutions are strongly committed to improving and harmonising document and information flows within the legislative process.

The adoption of a common format for the representation of the exchanged documents' content is a precondition to achieving this goal. This needs to be complemented by information flows based on a modern, robust and secure delivery mechanism to **replace the current e-mail exchanges**. Furthermore, every delivery needs to comprise **standardised metadata**, i.e. a structured and machine-readable description of the context and the content of the transmission. Together this will significantly increase interoperability on the technical level and it will allow for the automatisisation of low-level operations on the side of the recipient.

The definition of the **common format** for document content – driven by the European Institutions and facilitated by the Publications Office - takes place on two levels: **business and technical**. This documentation covers the technical level.

On the business level the "**Common Vocabulary**" (CoV) identifies and describes the semantic and syntactic concepts found in the exchanged documents. The collaborative analysis of commonly agreed examples will be extended step-by-step to cover all document types that are exchanged in the legislative process. The current release of the CoV (version 3.0) covers examples of regulations, directives, decisions and the proposals for these acts, as well as legislative resolutions of the European Parliament (without the consolidation of amendments). This release includes also the special mark-up of amending provisions and of simple patterns to be used in future annexes. The focus is on documents that are exchanged in the scope of the Ordinary Legislative Procedure (OLP), but an example of an international agreement has also been included.

The common understanding reached and documented in the CoV of the structure of the documents at business level is not sufficient to serve as specification for an implementation on technical level. In consequence, the technical level needs to be addressed by a model that transforms the definitions of the CoV into an implementable specification, which is referred to as "Akoma Ntoso for EU" (AKN4EU).

This documentation describes version 3.0 of AKN4EU.

The Institutions have agreed to use XML as a structured, machine-readable format for implementation and have chosen the Akoma Ntoso XML schema for the mark-up of the content. The current version of AKN4EU is based on Akoma Ntoso Version 1.0 OASIS Standard¹. **Akoma Ntoso, however, is a rather generic XML schema designed to cover a variety of legal documents. Consequently, it has to be specified how the semantic and syntactic concepts described by the CoV have to be expressed in Akoma Ntoso. In other words, AKN4EU can be considered as the localisation of Akoma Ntoso for EU legislation.**

¹ See https://www.oasis-open.org/committees/tc_home.php?wg_abbrev=legaldocml; namespace: <http://docs.oasis-open.org/legaldocml/ns/akn/3.0>.

AKN4EU is the key deliverable of the technical standardisation work and complemented by a configurable validation framework and a converter for XML legacy documents² to allow for the testing of the correctness of the specifications. The converter will also serve to support the testing of the implementations by the Institutions and, in particular, it will enable the reuse of existing legislation, or fragments thereof, by authors and other actors involved in the process.

These latter developments are not covered by this documentation.

² The converter converts the Publications Office's XML production format (Formex) into AKN4EU format (Akoma Ntoso).

1 Introduction

In the context of the Interinstitutional Metadata and Formats Committee (IMFC) the Institutions have decided to use XML as the structured, machine-readable format for implementation and have chosen the Akoma Ntoso XML schema for the mark-up of the content of the legislative documents managed and exchanged by the Institutions in the scope of the legislative process. The current version of AKN4EU is based on Akoma Ntoso Version 1.0 that has been accepted as an OASIS Standard.

As Akoma Ntoso is a rather generic XML schema that aims at covering various types of legal documents, it has to be specified how Akoma Ntoso should be applied to the different document types that are relevant for the legislative process at EU level.

These specifications comprise the **localization of Akoma Ntoso for EU** legislation and are referred to as **Akoma Ntoso for EU (AKN4EU)**.

This documentation introduces into the **working environment** (see Volume I Part 2 *Contextual issues to be considered* and Part 3 *Use cases*) and subsequently lays down in Part 4 the "**design principles**", i.e. the **principal decisions**, that form the core of AKN4EU, because they are applicable to all document types.

To establish and maintain the necessary collaboration with end users involved in drafting, translating or amending legislation, the Common Vocabulary (CoV) serves as a basis for the elaboration of AKN4EU. The CoV is developed at business level and provides in a human readable format a collection of semantic concepts that are needed to represent the content of the legislative documents. In a first phase, the CoV focuses on legislative documents pertaining to the Ordinary Legislative Procedure (OLP). AKN4EU defines the **mapping** of the CoV to Akoma Ntoso, i.e. it describes how the CoV concepts should be marked-up in Akoma Ntoso (see Volume II Part 2 *Correspondence between CoV and AKN4EU*). AKN4EU version 3.0 is based on CoV version 3.0³.

Part 1 in Volume II is an introductory chapter to AKN4EU providing guidance for the **XML mark-up of legislative documents of the EU**.

In **Volume II Part 3** the current definition of AKN4EU follows, providing detailed specifications on how the different Akoma Ntoso **tags and attributes** should be used in the context of EU documents pertaining to the OLP, illustrated by examples (full mark-up of existing documents is provided in separate folder 'Examples').

The definition of AKN4EU is complemented by physical specifications (**Volume I Part 6**), which comprise technical recommendations on the **character set, image file formats**, the exchange of data via IMMC and file naming conventions.

Human readable business rules are necessary to enrich the CoV and to complement AKN4EU, either where AKN4EU goes beyond the CoV and thus triggering a clarification need, or where implementation requires additional **business information** to be captured in order to know how to proceed. The approach for the **business validation rules is described in Volume I Part 7**.

³ See: CIRCABC > Publications Office > IMFC - Interinstitutional Formats and Metadata Committee Library > 05 CoV > Common Vocabulary

Finally, **Part 8 of** Volume I covers the technical validation rules that implement the business validation rules to ensure that they are respected for each exchange. The formal representation chosen per business rule might depend on the respective complexity or nature of the rule. Further technical validation rules focus on technical aspects, for example the file naming conventions.

This AKN4EU documentation is completed in Volume I **Part 9 with a glossary** also comprising an overview over the sources and references.

The comparison between the current and the previous version of the AKN4EU documentation is provided in form of a change log in the last part (Part 10) of this volume.

Important notes:

Progressively all exchanges of legal data between the Institutions should be based on AKN4EU. Therefore, AKN4EU has to be extended and updated on a regular basis to incorporate new requirements. Consequently, the documentation of AKN4EU constitutes a living document.

Most legislative documents are composed of a main document and one or several annexes. In order to serve as specification for implementation, AKN4EU also needs to cover the mark-up of the annexes. The Common Vocabulary (CoV) 3.0 covers selected patterns of annexes, which should become default patterns in the future (subject to approval by the GRITL, Groupe de réflexion sur la technique législative).

2 Contextual issues to be considered

The task of harmonizing the document exchanges pertaining to the legislative procedures between Institutions is challenging on multiple levels. The following issues need to be considered, preferably with a pro-active approach in applying the measures indicated below.

2.1 Better law-making

The greater context of better law-making comes with a variety of parallel projects and initiatives that are more or less related to the IMFC's work.

By investing some workload in the monitoring of such activities, OP aims at ensuring that relevant developments are considered and that, in return, other groups and initiatives are sufficiently informed of the IMFC's work (e.g. Groupe Interinstitutionnel sur la Technique Législative (GRITL), Interinstitutional Interoperability Steering Committee (IISC), Interinstitutional Informatics Committee (CII), OP Management Committee).

2.2 OLP only first focus

The IMFC's mandate focuses on documents pertaining to the OLP. It comes with the nature of the Institutions' developments in the field that other documents and activities play an important role and are requested to be considered.

The IMFC responds to this need for flexibility by taking on board newly proposed document types (e.g. "International Agreement") and by liaising with ongoing and closely related developments (e.g. on the Trilogue Table Editor (TTE), EdiT).

2.3 Avoiding a loss of momentum

The setting up of the IMFC created some momentum between the participating Institutions, also based on initial consensus and commitment to the common goal. To meet the risk of losing this momentum, OP has made sure that a frequent report to middle (IMFC Plenary) and senior management (OP Management Committee) also translates down to the Institutions' contributing on working group level.

2.4 Parallel developments

2.4.1 Need for alignment and interoperability

In parallel to the harmonization at interinstitutional level, the participating Institutions are facing the challenge of advancing the work on their internal systems in the related field (for example, the European Parliament and the Council working on their Akoma Ntoso implementation, development of the Trilogue Table Editor or of a common editing tool, EdiT, by the Council and the Commission). In this regard, the Interinstitutional Interoperability Steering Committee provides guiding principles such as for the reuse of technology and best practices, the convergence of technical solutions considering the Institutions' particularities and IT context, the need to apply common standards and to ensure compatibility of formats, as well as the respect of multilingualism.

The IMFC aims at benefiting from these developments by involving the key players as far as possible to assure consistency in the approach and compatibility of the solutions.

2.4.2 Additional workload – limited resources

The Institutions' contribution to the IMFC's work represents additional workload. As this impacts the Institutions' key players who are also leading the internal projects, this additional demand meets particularly limited resources that are, by the nature of the work, highly qualified and specialized.

OP meets this challenge by frequently reminding the management to raise awareness for the colliding high priority interests so that the necessary resources are nevertheless sufficiently available, in particular in key phases for the deliverables.

2.4.3 Developing with conflicting requirements

The focus of the ongoing implementation work lies on the development of appropriate editing tools for authors. The context and requirements for such developments can cause conflict:

- The adoption of AKN4EU may require changes in ongoing implementations, which can result in delays compared to an initial planning. Conflicting implementation interests must not negatively affect the substance and the consistency of AKN4EU.
- The mark-up supported by the editing tools might not be sufficient to meet the requirements for all use cases (translation, publishing...). One challenge comes with finding the compromise between the desired level of flexibility for authors and the need for reliable structural information in other contexts.

2.5 Impact on internal and external end users

The impact on the different categories of internal and external end users' needs to be anticipated, assessed and addressed. From the institutions' point of view, and considering their technological perspective, this should comprise among others:

- Drafters and translators: what tools can support colleagues involved in the processes to facilitate a smooth transition to, and maximum benefit from, the common format?
- Member States and other institutions: how will use cases be handled such as “citing a provision”, “interpreting it in a judgment of the CJEU”, “transposing it in national law” or “handling amendments and consolidation”?
- Citizens: how will the use of a common format, based on XML, improve accessibility and reusability of the EU law? How will this improve transparency?
- Taxpayers: what is the economic impact of the reduction of conversions?

Remark:

XML structures allow for more flexibility and can facilitate the support of the drafters during the drafting process by exploiting the advantages that automation offers. For example, a part of a document can easily be extracted by external tools for further processing (translation or amendment purposes).

2.6 Overall stakeholder acceptance

The overall success of the IMFC's work for the Institutions will strongly depend on acceptance by stakeholders at all levels, including end users (for example if it comes to changing the daily working environment by introducing tailored solutions for the editing of XML documents).

The Institutions' parallel projects run under the very same pre-condition, which facilitates the acceptance of the interinstitutional approach.

In addition, for the IMFC's work, the risk of an unacceptably low level of acceptance by the end users would only become apparent at a late stage. The necessary preparations to combat this (raising awareness, training, testing) need to be tackled at a sufficiently early stage across the Institutions.

3 Use cases

The IMFC's focuses primarily on documents pertaining to the OLP and on the use case 'exchange of documents'⁴. As regards the latter, the IMFC's approach and work need to take into account the activities in the metadata context as well as the results of the 'AS-IS' and 'TO-BE' landscaping exercises.

As a starting point, the IMFC had approved a list of document types to be analysed as a basis for the CoV, and subsequently for AKN4EU. The list of example documents covered so far can be found in section 5 'Examples'.

In addition, and in order to conform with the Institutions' requirements and user expectations, the general concepts of reuse and format conversion need to be put into perspective regarding the following Institutions'/end user tasks at hand, which all concern document content:

- drafting
- translating
- proof-reading
- revising
- amending
- adopting
- publishing
- citing/referencing⁵
- correcting/corrigenda
- consolidating/codifying/recasting.

⁴See CIRCABC > Publications Office > IMFC > Library > 00 IMFC mandate.

⁵The "citing/referencing" use-case could be enhanced by the support of "synonyms" between dispositions of EU law (to support the case when "article X of directive A" becomes "article Y of directive B", which would imply a specific mark-up of the former dispositions when they are indicated)

4 Design principles

The design principles constitute the basic specification covering aspects common to all document types.

4.1 Compliance with CoV

AKN4EU has to be compliant with the CoV.

For each CoV concept the corresponding representation in Akoma Ntoso has to be defined.

AKN4EU has to respect the business rules that have been defined in the context of the CoV.

AKN4EU also includes concepts which are not relevant at business level, but which are necessary for the technical implementation. These concepts are by their nature not defined in the CoV.

4.2 Synoptism and granularity of multilingual parallel texts

One particularity of the European Union's legislative context is the **legal obligation to represent legislation in all official languages of the EU**. Texts that are representations of the same content in different languages could be defined as multilingual parallel texts⁶.

For the multilingual parallel texts based on AKN4EU, the building blocks that comprise the XML tree of each individual text (or document) should be in principal identical across all linguistic versions.

This characteristic is also called synoptism.

Synoptism between different linguistic versions can only be guaranteed down to a certain level of detail because of differences that are imposed by language specific rules and conventions.

The level of detail, up to which text segments can automatically be aligned, is referred to as *parallel text granularity*⁷**Error! Bookmark not defined..**

In the scope of AKN4EU, the lowest level of parallel text granularity is a **paragraph** (block of text).

Synoptism improves the efficiency and quality of the document production processes in a multilingual environment and facilitates the structural comparability across the various linguistic versions.

Remark:

A finer granularity may be possible: a paragraph can comprise inline elements such as e.g. dates, numbers or references. However, even if these elements are usually language independent, the order in which they appear is language dependent.

The system should be aware of what the smallest block for synoptic purposes is at any given point of the content (it could be one of a variety of elements, e.g. a title, a (sub)paragraph or a list, etc.).

⁷ See <https://arxiv.org/ftp/arxiv/papers/0808/0808.3889.pdf>.

4.3 Annexes

One main document (in general an act) may be accompanied by one or multiple annexes.

Annexes are considered by their nature as **integral part of a document** and can be essential for the understanding.

The content of an annex may follow various structures, because there have been no specific rules governing the structure of annexes. Nevertheless, some common patterns have been identified and designated as models for drafting of annexes in the future.

The main document itself constitutes an XML document stored as a separate file and contains links to its respective annex(es). Each annex itself constitutes also an XML document stored as a separate file.

An annex can contain appendices. In the same way as the main document contains links to its respective annex(es), an annex contains links to its appendices. Each appendix itself constitutes also an XML document stored as a separate file.

One advantage of the separation of the act and its annexes is the possibility to provide, if required, each document individually, instead of being limited to present the collection of all components, for example when local updates affect only one annex to a document. A mechanism of transport of an individual XML file will be defined.

4.4 Principles for the definition of identifiers and references

Identification and reference systems are essential when it comes to re-using or exchanging documents between the Institutions. It is a **precondition for successful collaboration that identifiers at document level are preserved throughout any exchange**.

The different types of references (e.g. legal basis; universal references or references common to all Institutions) need to be identified in the CoV.

The **European Legislation Identifier (ELI)⁸** shall be used for the representation of references wherever possible.

The naming convention of ELI is of considerable importance because it supports the concepts *Work*, *Expression* and *Manifestation*, as defined by the FRBR model⁹, which is used inside the IMMC envelope and inside Akoma Ntoso documents.

The ELI naming convention for documents shall be adopted by AKN4EU. For document types not yet covered by ELI, a draft proposal is prepared and submitted to the ELI Project team for validation and approval. For example, the following ELI identifiers are draft proposals for document types currently not covered by ELI, and are still under revision by the ELI taskforce:

- Proposal for an act, COM(2016) 53
ELI identifier: http://data.europa.eu/eli/prop_act/2016/53

⁸ See <https://publications.europa.eu/en/web/eu-vocabularies/eli>

⁹ See <https://www.loc.gov/cds/downloads/FRBR.PDF>

- Commission Opinion on the amendments made by the European Parliament, COM(2011) 179
ELI identifier: http://data.europa.eu/eli/opin_amend_ep/2011/179
- Communication concerning the position of the Council, COM(2017) 127:
ELI identifier: http://data.europa.eu/eli/communic_posit/2017/127
- Commission staff working document with impact assessment, SWD(2016)168:
ELI identifier: <http://data.europa.eu/eli/swd/2016/168>
- Commission staff working document with executive summary of the impact assessment, SWD(2016)169
ELI identifier: <http://data.europa.eu/eli/swd/2016/169>
- Commission staff working document, SWD(2016)170
ELI identifier: <http://data.europa.eu/eli/swd/2016/170>.

For the draft proposals above, the naming convention of the document type itself is based on the common [authority table Resource type](#)¹⁰.

ELI is also used for the definition of references inside a document, i.e. URIs of individual structural elements (for example for the definition of the URI of an article). Such URIs will allow direct referring to the part of a text that is cited and will facilitate the re-use of parts (e.g. articles) of an existing document during the drafting phase of new acts. These identifiers can also be used to control *synoptism* across linguistic versions and to improve the structural quality of documents' content. Furthermore, the identifiers can facilitate the support of “synonyms” or “replacement” of former dispositions by new ones (for instance in correlation tables, sometimes provided as annexes, e.g., see Annex III of [32017R1001](#)).

A ‘functional reference’ to the structural elements inside a document is based on the structure of the document as defined by the CoV.

A ‘technical reference’ is based on the *xml:id* attribute (the AKN4EU technical identifier).

4.5 Reference to authority tables

The EU Vocabularies website¹¹ provides access to the common vocabularies managed by the Publications Office on behalf of the EU institutions and bodies. This includes the Metadata Registry (MDR), the EuroVoc thesaurus, business collections, etc. The MDR registers and maintains definition data (metadata elements, Named Authority Lists (NALs), schemas, ontologies, etc.).

The institutions agreed to rely on these common vocabularies in the scope of AKN4EU to standardize attributes and to increase interoperability.

Codes from the Named Authority Lists¹² (also known as Common Authority Tables) shall be used whenever possible, i.e. on the level of the IMMC envelope and inside the document. Whenever necessary, new codes should be proposed and added to the respective NAL according to the MDR governance procedure.

¹⁰<https://publications.europa.eu/en/web/eu-vocabularies/at-dataset/-/resource/dataset/resource-type>

¹¹<https://publications.europa.eu/en/web/eu-vocabularies/home>

¹²<https://publications.europa.eu/en/web/eu-vocabularies/authority-tables>

As Akoma Ntoso requires the use of URI for referencing metadata, the common way of referencing metadata stored in the authority tables is to use the URI defined by the Cellar, the Publications Office's common repository for metadata and content. For example, the following URI from the authority table *Corporate body* is used for referencing the term "European Commission" in AKN4EU documents:

<http://publications.europa.eu/resource/authority/corporate-body/COM>

The authority tables also allow a mapping to facilitate a different use of the codes depending on the need. For example, the two-letter code in lowercase used as language code in AKN4EU has to be mapped with the three-letter authority code in uppercase.

4.6 Mark-up of tables

Tables should be marked up according to Akoma Ntoso (use of the standard HTML table model). For complex cases, which cannot be marked-up using the HTML table model, images should be used.

4.7 Packaging of XML files

A single manifestation of a document can comprise multiple XML files. For each linguistic version, they are bundled in a zip file with the extension called "leg" (short for: legislation).

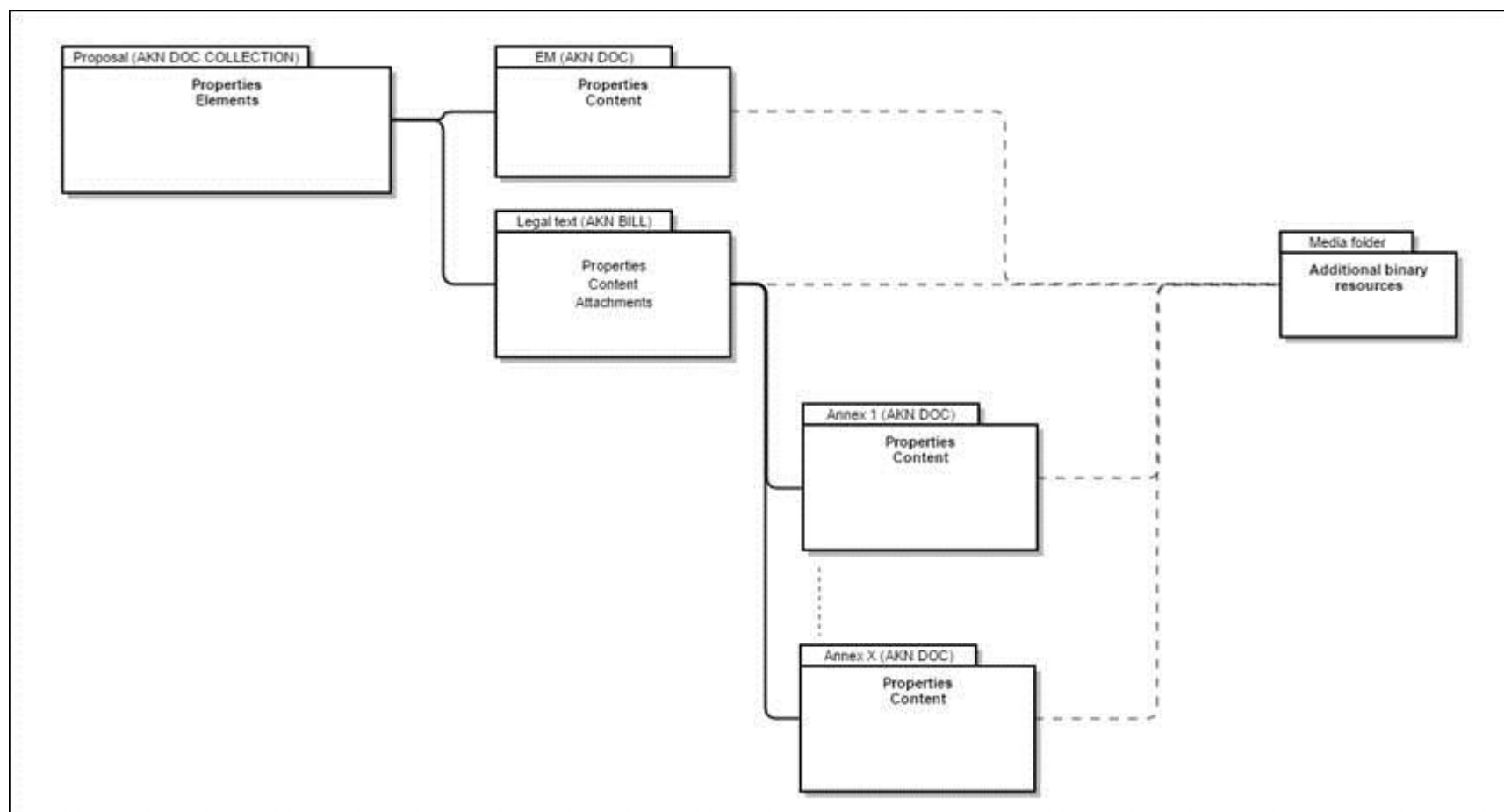
It is necessary that the .leg zip file exists not only for the whole package, but also for each of the documents it contains separately.

The Commission has provided an example of how the bundle of XML files that constitute a legislative proposal is represented in their implementation. It has not yet been decided whether AKN4EU will use exactly the same structure.

Key features of the example representation for the proposal are the following:

- The entry point is a file with a standardized name (*main.xml*), containing the global proposal (for the proposal it is a *documentCollection*).
- Each component, which is an Akoma Ntoso document type, is stored in a specific xml file. This xml file is referenced in the document, to which it is a sub-document.
- Therefore, the *documentCollection* references:
 - the explanatory memorandum (a *doc*), and
 - the legal text (a *bill*). The bill may contain one or more annex(es).

The figure below shows the Commission's representation of a legislative proposal ("EM" stands for Explanatory Memorandum). A transformation process can easily extract the bill so that it can be provided to the Council and/or the Parliament as a valid Akoma Ntoso representation of a proposal, ready to be easily amended.



4.8 Relation between AKN4EU and IMMC

AKN4EU documents will be exchanged between Institutions using the IMMC protocol.

Each document needs to be self-contained: it constitutes a complete and stand-alone entity that can be used independently of its IMMC metadata. To benefit from this, the metadata contained in the metadata section of the document are duplicated in the IMMC envelope as necessary.

As a result, no link is required between the AKN4EU instance and the IMMC file.

To illustrate this, examples of IMMC packages/containers are presented in section **Physical specifications** of the documentation.

The Akoma Ntoso schema (Akoma Ntoso) requires that the definition of the metadata section of AKN4EU takes into account some mandatory Akoma Ntoso metadata elements.

4.9 Best practices

The Institutions agreed on the following as best practice:

- Iterative approach:

The definition of AKN4EU will follow an iterative approach, i.e. the mark-up will be enhanced progressively.

- Proprietary mark-up:

Information that is proprietary to one Institution should be preserved throughout the exchanges. The feasibility of preserving proprietary information during exchanges will be handled on a case-by-case basis. The tagging of this information with a specific mark-up will guarantee the preservation for the sender and will facilitate the processing for the recipient.

- Focus on structure:

There is consensus that the focus needs to be on structure. Presentational aspects shall not be a priority.

5 Examples

The list of example documents selected for analysis, and covered by CoV 3.0¹³ and AKN4EU 3.0, is presented below. The instances are provided in the folder ‘Examples’ of the delivery package (temporary repository). By default, for each example document there is one CoV and one AKN4EU instance, unless indicated otherwise below.

- CoV_DLV16_1.1 32014R1384
Regulation (EU) No 1384/2014 of the European Parliament and of the Council of 18 December 2014 on the tariff treatment for goods originating in Ecuador
- CoV_DLV17_1.2 52016PC0818
Proposal for a REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL amending Regulation (EC) No 1008/2008 on common rules for the operation of air services in the Community
 - example of IMMC package (the ‘leg’ file and the corresponding IMMC instance):
 proposal-COM2016818_2016-07-19T161847528_immc.zip
 - AKN4EU instances composing the proposal:
 - main.xml: the entry point of the proposal,
 - memorandum_123.xml : the explanatory memorandum,
 - bill_123.xml: the bill of the proposal,
 - XML package as ‘leg’ file containing all instances:
 CoV_DLV17_1.02.PROP_REG.52016PC0818.akn4eu3.0.20191212.leg
- CoV_DLV17_1.5 32013D1313
Decision No 1313/2013/EU of the European Parliament and of the Council of 17 December 2013 on a Union Civil Protection Mechanism
- CoV_DLV17_1.6 32013R1380
Regulation (EU) No 1380/2013 of the European Parliament and of the Council of 11 December 2013 on the Common Fisheries Policy, amending Council Regulations (EC) No 1954/2003 and (EC) No 1224/2009 and repealing Council Regulations (EC) No 2371/2002 and (EC) No 639/2004 and Council Decision 2004/585/EC
- CoV_DLV17_1.7 32015L0254
Directive (EU) 2015/254 of the European Parliament and of the Council of 11 February 2015 repealing Council Directive 93/5/EEC on assistance to the Commission and cooperation by the Member States in the scientific examination of questions relating to food
 - example of IMMC package (one ‘leg’ file and the corresponding IMMC instance): act-55698743285_2017-06-12T073151577_immc.zip
- CoV_DLV18_1.1 52013PC0824

¹³ The example annexes listed were selected to illustrate priority patterns defined by the Institutions.

Proposal for a DIRECTIVE OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL on provisional legal aid for suspects or accused persons deprived of liberty and legal aid in European arrest warrant proceedings

- CoV_DLV18_1.2 52016PC0053
Proposal for a DECISION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL on establishing an information exchange mechanism with regard to intergovernmental agreements and non-binding instruments between Member States and third countries in the field of energy and repealing Decision No 994/2012/EU
 - AKN4EU instances composing the proposal:
 - main.xml: the entry point of the proposal,
 - memorandum_123.xml : the explanatory memorandum,
 - bill_123.xml : the bill of the proposal,
 - XML package as 'leg' file containing all instances:
CoV_DLV18_1.02.PROP_DEC.52016PC0053.akn4eu3.0.20191212.leg
- CoV_DLV18_1.5 P8_TA-PROV(2017)0425 (only the legislative resolution part)
European Parliament legislative resolution of 14 November 2017 on the proposal for a directive of the European Parliament and of the Council on the recognition of professional qualifications in inland navigation and repealing Council Directive 96/50/EC and Council Directive 91/672/EEC
- CoV_DLV18_1.6 32013R1291
Regulation (EU) No 1291/2013 of the European Parliament and of the Council of 11 December 2013 establishing Horizon 2020 - the Framework Programme for Research and Innovation (2014-2020) and repealing Decision No 1982/2006/EC
- CoV_DLV18_1.7 32017D0684
Decision (EU) 2017/684 of the European Parliament and of the Council of 5 April 2017 on establishing an information exchange mechanism with regard to intergovernmental agreements and non-binding instruments between Member States and third countries in the field of energy, and repealing Decision No 994/2012/EU
- CoV_DLV18_1.8 32014R0254
Regulation (EU) No 254/2014 of the European Parliament and of the Council of 26 February 2014 on a multiannual consumer programme for the years 2014-20 and repealing Decision No 1926/2006/EC
- CoV_DLV18_1.11 22016A1203(03)
Strategic Partnership Agreement between the European Union and its Member States, of the one part, and Canada, of the other part
- CoV_DLV18_1.12 32017R2393
Regulation (EU) 2017/2393 of the European Parliament and of the Council of 13 December 2017 amending Regulations (EU) No 1305/2013 on support for rural development by the European Agricultural Fund for Rural Development (EAFRD), (EU) No 1306/2013 on the financing, management and monitoring of the common agricultural policy, (EU) No 1307/2013 establishing rules for direct payments to farmers under support schemes within the framework of the common agricultural policy, (EU) No 1308/2013 establishing a common organisation of the

markets in agricultural products and (EU) No 652/2014 laying down provisions for the management of expenditure relating to the food chain, animal health and animal welfare, and relating to plant health and plant reproductive material

- CoV_DLV18_1.16 32017L1564
Directive (EU) 2017/1564 of the European Parliament and of the Council of 13 September 2017 on certain permitted uses of certain works and other subject matter protected by copyright and related rights for the benefit of persons who are blind, visually impaired or otherwise print-disabled and amending Directive 2001/29/EC on the harmonisation of certain aspects of copyright and related rights in the information society
- CoV_DLV19_1.1 32017R0625 (only annex)
Annex II to Regulation (EU) 2017/625 of the European Parliament and of the Council of 15 March 2017 on official controls and other official activities performed to ensure the application of food and feed law, rules on animal health and welfare, plant health and plant protection products, amending Regulations (EC) No 999/2001, (EC) No 396/2005, (EC) No 1069/2009, (EC) No 1107/2009, (EU) No 1151/2012, (EU) No 652/2014, (EU) 2016/429 and (EU) 2016/2031 of the European Parliament and of the Council, Council Regulations (EC) No 1/2005 and (EC) No 1099/2009 and Council Directives 98/58/EC, 1999/74/EC, 2007/43/EC, 2008/119/EC and 2008/120/EC, and repealing Regulations (EC) No 854/2004 and (EC) No 882/2004 of the European Parliament and of the Council, Council Directives 89/608/EEC, 89/662/EEC, 90/425/EEC, 91/496/EEC, 96/23/EC, 96/93/EC and 97/78/EC and Council Decision 92/438/EEC (Official Controls Regulation)
- CoV_DLV19_1.2 32017R745 (only annex)
Annex XVI to Regulation (EU) 2017/745 of the European Parliament and of the Council of 5 April 2017 on medical devices, amending Directive 2001/83/EC, Regulation (EC) No 178/2002 and Regulation (EC) No 1223/2009 and repealing Council Directives 90/385/EEC and 93/42/EEC
- CoV_DLV19_1.5 32014R0223 (only annex)
Annex II to Regulation (EU) No 223/2014 of the European Parliament and of the Council of 11 March 2014 on the Fund for European Aid to the Most Deprived
- CoV_DLV19_1.6 32017L2397 (only annex)
Annex IV to Directive (EU) 2017/2397 of the European Parliament and of the Council of 12 December 2017 on the recognition of professional qualifications in inland navigation and repealing Council Directives 91/672/EEC and 96/50/EC
- CoV_DLV19_1.14 32018R0858 (only annex)
Annex IX to Regulation (EU) 2018/858 of the European Parliament and of the Council of 30 May 2018 on the approval and market surveillance of motor vehicles and their trailers, and of systems, components and separate technical units intended for such vehicles, amending Regulations (EC) No 715/2007 and (EC) No 595/2009 and repealing Directive 2007/46/EC
- CoV_DLV19_1.15 32017R0852 (only annex)
Annex IV to Regulation (EU) 2017/852 of the European Parliament and of the Council of 17 May 2017 on mercury, and repealing Regulation (EC) No 1102/2008
- CoV_DLV19_1.16 32017R0746 (only annex)

Annex II to Regulation (EU) 2017/746 of the European Parliament and of the Council of 5 April 2017 on in vitro diagnostic medical devices and repealing Directive 98/79/EC and Commission Decision 2010/227/EU

- CoV_DLV19_1.17 32016R0425 (only annex)
Annex II to Regulation (EU) 2017/746 of the European Parliament and of the Council of 5 April 2017 on in vitro diagnostic medical devices and repealing Directive 98/79/EC and Commission Decision 2010/227/EU
- CoV_DLV19_1.18 32017L2103 (only annex)
Annex II to Regulation (EU) 2017/746 of the European Parliament and of the Council of 5 April 2017 on in vitro diagnostic medical devices and repealing Directive 98/79/EC and Commission Decision 2010/227/EU
- CoV_DLV19_1.19 32017R2395
Regulation (EU) 2017/2395 of the European Parliament and of the Council of 12 December 2017 amending Regulation (EU) No 575/2013 as regards transitional arrangements for mitigating the impact of the introduction of IFRS 9 on own funds and for the large exposures treatment of certain public sector exposures denominated in the domestic currency of any Member State¹⁴
- CoV_DLV19_1.20 32019R1241 (only annex)
Annex IV to Regulation (EU) 2019/1241 of the European Parliament and of the Council of 20 June 2019 on the conservation of fisheries resources and the protection of marine ecosystems through technical measures, amending Council Regulations (EC) No 1967/2006, (EC) No 1224/2009 and Regulations (EU) No 1380/2013, (EU) 2016/1139, (EU) 2018/973, (EU) 2019/472 and (EU) 2019/1022 of the European Parliament and of the Council, and repealing Council Regulations (EC) No 894/97, (EC) No 850/98, (EC) No 2549/2000, (EC) No 254/2002, (EC) No 812/2004 and (EC) No 2187/2005

¹⁴ Under discussion.

6 Physical specifications

The physical specifications comprise technical recommendations concerning:

1. the character set,
2. the format of images,
3. the exchange of data and
4. the construction of filenames.

For illustration purposes, some examples are added.

6.1 Character encoding

The required character encoding to be used in Akoma Ntoso documents is based on UTF-8 (8-bit Unicode Transformation Format).

The encoding covers the official languages of the European Union.

Character entities:

Character entities may only be used under the following circumstances:

- numeric values corresponding to the Unicode code such as ' ' for 'no-break-space';
- numeric values for 8-bit Unicode, e.g. Chinese names;
- special characters having a specific function within an XML instance: '<', '>', '"', '&' and '''.

Character range:

The range of Unicode characters allowed in the documents is the following:

0020-007E, 00A0-00A3, 00A5-00A7, 00A9, 00AB-00AC, 00AE, 00B0-00B1, 00B5, 00B7, 00BB-00FF, 0100-0107, 010A-0113, 0116-011B, 011E-0123, 0126-0127, 012A-012B, 012E-0133, 0136-0137, 0139-0148, 014A-014D, 0150-015B, 015E-0165, 016A-016B, 016E-0173, 0178-017E, 0192, 01C4-01CC, 0218-021B, 0374, 037E, 0386-038A, 038C, 038E-03A1, 03A3-03CE, 0401-0405, 0407-042A, 042C, 042E-044A, 044C, 044E-044F, 0451-0455, 0457-045F, 0490-0491, 2010, 2013-2014, 2018-2022, 2026, 2030, 2032-2034, 2039-203A, 203E, 20AC, 2116, 2122, 2153-2154, 215B-215E, 2190-2199, 21D0-21D9, 21E6-21E9, 2213, 221A, 221E, 222B, 2243, 2245, 2248, 2260-2267, 2276-2277, 227C-227D, 22D5, 22DE-22DF, 2300, 2326, 2329-232B, 25A0, 25C7, 2605-2606, 2640, 2642, F106-F10A, F1B1, FB01-FB02.

For characters from right-to-left languages, such as Arabic and Hebraic, both images and characters are accepted but it is strongly recommended to use characters instead of images. If an image is used, the corresponding characters must also be encoded.

Further details regarding the character encoding in XML documents are described in the physical specifications of Formex (Formalized Exchange of Electronic Publications) [<http://formex.publications.europa.eu/formex-4/physspec/formex-4-character-encoding-index.htm>].

The encoding includes rules concerning the range of characters allowed per language and also quotation marks, punctuation, parenthesis, arrows, currency symbols, mathematical operators and symbols, digits as well as Cyrillic, Greek and Latin letters.

Information on the use of special characters can also be found in the "Interinstitutional Style Guide" [<http://publications.europa.eu/code/en/en-000300.htm>].

6.2 Image formats

The following image formats are supported in Akoma Ntoso documents:

- base64 images (support of based64 as this is used by the European Commission for proposals; the attribute **src** starts with "data:image/png;base64,"),
- images as external files (support of independent files as this is used by the European Parliament).

Remark:

Setting a list of allowed formats is important not only for defining the rules for an automatic validation, but also to allow the conversion towards other formats as well as for long-term preservation.

In the production workflow of the Publications Office, all images received are converted to TIFF format. The resolution is set to 400 dpi. (By definition, the resolution is always present in any image.) Further technical requirements concerning the TIFF format are described in the physical specifications of Formex (Formalized Exchange of Electronic Publications) [<http://formex.publications.europa.eu/formex-4/physspec/formex-4-TIFF.htm>].

List of allowed standards (to be completed and the use of features and standards internal to the formats are to be defined):

- PNG
- SVG

6.3 Container format – integration with IMMC

The XML instances will be exchanged through IMMC packages.

By definition, an IMMC package is a zip file which contains the XML instances that comprise one (or more) linguistic version(s) of a document, and are accompanied by the corresponding IMMC instance.

The following example (of an IMMC package for a “Proposal for a regulation”, English language) illustrates the structure of a full IMMC package and shows how the XML instances are referred to in IMMC:

IMMC package zip file:

- The leg file [zip file: extension '.leg']:
 - main.xml: the starting point. Contains the documentCollection with the main components declaration of the proposal.
 - memorandum_<identifier>.xml: content of the explanatory memorandum, in xml.
 - bill_<identifier>.xml: content of the bill, in xml.
- The IMMC instance: as currently defined within the Commission, in xml.

The example IMMC package is available in the folder 'examples/immc'.

It comprises the 'Proposal for a Regulation of the European Parliament and of the Council amending Regulation (EC) No 1008/2008 on common rules for the operation of air services in the Community' (CoV DLV17 1.2, CELEX number: 52016PC0818).

6.4 File naming conventions

- The Institutions agreed on the minimum requirement that the filename of XML instances and packages **must not contain space characters**.
- An XML package of XML instances that comprises a document carries the file name extension '.leg'.
- XML instances within a package of a **proposal** have the following file names:
 - main.xml: the documentCollection with the main components declaration of the proposal;
 - memorandum_<identifier>.xml: the explanatory memorandum;
 - bill_<identifier>.xml: the bill;
 - annex_<identifier>.xml: the annex(es) to the bill, if any. One separate file per annex. Appendices should be treated as annexes, each in separate file.
- XML instances within a package of an **act** have the following file names:
 - main.xml: the act.
 - annex_<identifier>.xml: the annex(es), if any. One separate file per annex. Appendices should be treated as annexes, each in separate file.
- An IMMC package has the file name extension 'zip'.

Based on the IMMC Council transmission protocol rules specifications, the naming convention used for the file transmission between the Council and the Publications Office are as follows:

<work_identifier>_<date_of_transmission>T<timestamp_of_transmission>

where:

- date_of_transmission follows the pattern yyyy-mm-dd and
- timestamp_of_transmission follows the pattern HHMMSSmmm (Hours, Minutes, Seconds, milliseconds).

Currently there are no specific requirements defined for the naming convention to be applied for the file transmission between the European Commission and the European Parliament.

7 Business validation rules

This chapter describes in a human readable way the business validation rules needed for the definition of AKN4EU (formerly CEM). Each business rule shall have a unique identifier (e.g. AKN4EU_BUS_<nnnn> where n is a digit).

The basic processing of XML documents needs to comprise the parsing to guarantee that documents are well-formed (assuring that basic XML requirements are met, for example that every opening tag has a matching closing tag).

The next level requirement of conforming to a given schema needs to be met by a respective validation against that schema.

Some of the concepts defined in the Common Vocabulary need to be enriched by additional information, be it for AKN4EU to go beyond the concept definition by specifying additional rules, be it by the CoV adding necessary information for the implementation of a given concept.

An example for AKN4EU going beyond the concepts defined in the CoV:

The CoV does not see the need to specify a specific concept for the list of points. AKN4EU needs to specify a concept that gathers the "points" (of a list) to be able to address this higher level entity.

8 Technical validation rules

(This chapter comprises work in progress and will be updated regularly to incorporate new requirements.)

In order to ensure that each exchange based on AKN4EU complies with the business rules defined, every human readable business rule has to be translated into a formal representation. There might be different implementations possible, depending on the nature or the complexity of each business rule. There will be rules, which will already be verified by the parsing of the individual XML instances. Other rules could be described by Schematron and complex rules could even require the development of a dedicated validation module.

In addition, there will be technical validations that are not related to business rules. Examples are the **checking of file naming conventions, of character sets or of data types (e.g. date)**.

Subsequently, this chapter will provide the following:

- A correspondence between business and technical validation rules (documentation);
- A formal description of technical validation rules (formal representation) (the work on technical validation rules is ongoing);
- The documentation of the technical validation rules, supported by examples (drafting of the documentation is ongoing);
- The definition of compliance levels;
- Schematron rules (a Schematron tool has been developed and is updated regularly based on the draft documentation of the technical validation rules);
- The technical specifications for prototypes, in particular for validation of complex technical validation rules.

9 Glossary

Acronym	Definition
IMFC	Interinstitutional Metadata and Formats Committee. It manages the elaboration and maintenance of common interinstitutional standards for the exchange of legal data (content and metadata). The IMFC can establish subgroups for specific topics for a limited period of time. It can also establish up to two permanent subgroups for the coordination on the operational and technical level, which includes both subgroups “Common Vocabulary” and “Formats Guidelines”.
CII	Interinstitutional Informatics Committee
IISC	Interinstitutional Interoperability Steering Committee
GRITL	Groupe Interinstitutionnel sur la Technique Législative
OLP	Ordinary legislative procedure
TTE	Trilogue Table Editor
EdiT	Editing tool under development, to be used by the Council and the Commission.
CoV	Common Vocabulary
CEM	Common Exchange Model (re-named to AKN4EU)
AKN	Akoma Ntoso, OASIS standard. [https://www.oasis-open.org/committees/tc_home.php?wg_abbrev=legaldocml] [http://docs.oasis-open.org/legaldocml/ns/akn/3.0] Akoma Ntoso Version 1.0 and Akoma Ntoso Naming Convention Version 1.0 have been approved as OASIS standard in September 2018..
AKN4EU	Application of Akoma Ntoso for the mark-up of EU documents (formerly: CEM).
MDR	Metadata Registry accessible via the EU Vocabularies website. [https://publications.europa.eu/en/web/eu-vocabularies/home]
FRBR	Functional Requirements for Bibliographic Records [https://www.loc.gov/catdir/cpsd/whatfrbr.html]
IMMC	Interinstitutional Metadata Maintenance Committee [https://publications.europa.eu/en/web/eu-vocabularies/immc]
NALs	Named Authority Lists, also known as Common Authority Tables. [https://publications.europa.eu/en/web/eu-vocabularies/authority-tables]
ELI	European Legislation Identifier [https://publications.europa.eu/en/web/eu-vocabularies/eli]
Formex	Formalized Exchange of Electronic Publications [http://formex.publications.europa.eu/index.html]
leg	Legislation (leg is the extension file used by the Commission for the example package of XML files)
XML	eXtensible Markup Language [https://www.w3.org/XML/]
UTF-8	8-bit Unicode Transformation Format [https://en.wikipedia.org/wiki/UTF-8]
URI	Uniform Resource Identifier [https://en.wikipedia.org/wiki/Uniform_Resource_Identifier]
IRI	Internationalized Resource Identifier [https://en.wikipedia.org/wiki/Internationalized_Resource_Identifier]

dpi	Dots per inch [https://en.wikipedia.org/wiki/Dots_per_inch]
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10 Change Log: AKN4EU version 2.1 and AKN4EU version 3.0

(section to be updated)

This part lists the most important updates of AKN4EU version 3.0. The changes are indicated in the table below, where each row represents a relevant change, indicating the section number it is found in (for both versions), and the type of change (i.e. new, updated, moved or deleted). Minor updates (i.e. spelling corrections or formatting issues) are not mentioned.

AKN4EU version 2.1	AKN4EU version 3.0	Type of change	Details
Parts 2 – 4 and Parts 6 - 9		updated	See Tracked changes.
5. Examples		updated	Added new example documents in the list.
Volume II Part 1 AKN4EU Guidelines		updated	See chapters highlighted in green and Tracked changes
Volume II Part 2 AKN4EU Specifications	Volume II Part 3 AKN4EU Specifications	updated	Order of parts changed. See Tracked changes.
Volume II Part 3 Correspondence between CoV and AKN4EU	Volume II Part 2 Correspondence between CoV and AKN4EU	updated	Order of parts changed.